Exercise 6: An anagram finder

An anagram is a word formed by rearranging the letters of another word. We can say one word is an anagram of another if both have the same number of letters (for instance, "cat" and "tar", which both have one "c", one "a", and one "t"). Let's make a program which can find _all_ the anagrams in a given file list of words.

1. First of all, download the list of words "words.txt" from LC. Notice that each word in the file is on a new line. Write code which will load "words.txt" so that each word is an element of a list, and any newline character at the end of the word is removed. Add code which will search this list for the word "data", and print out the index of this word.

NOTE: In order to load the file in Colab, you'll need to add it to your Files list (from the left-hand navigation pane). Go to your list of files, and choose the "Upload" option. Once uploaded, you'll be able to work with your file from within Python as normal. Colab will warn you that the file storage is temporary (you'll have to re-upload it again if you return to your notebook), but that's fine for now (permanent storage should be on a Google Drive, for example).

2. Write code which will iterate over all the words in your list, and use a Counter (or a dictionary, if you prefer) for each to build a histogram of the number of letters in that word (print the word and the resulting counter out each time, so that you can see the results). Now, add a new dictionary, and store your counter for each word in that dictionary (with the word as the key, and the Counter as the value).

3. Next, write code which takes your dictionary of words : letter_counts, and creates a new dictionary, this time with a single counter per key, and a list of words which match that counter as its value.

4. Add code which iterates over your finished dictionary, and counts the total number of anagrams found. How many are there in the whole of "words.txt"?

You might do this by simply iterating over the dictionary and noticing where a set of letter counts are related to more than one word, or by iterating over it and deleting any entry which only has one word associated with it (then finding out how many keys remain).

You could even iterate over your list of anagrams, and print out each group, so you can see what your code discovered!